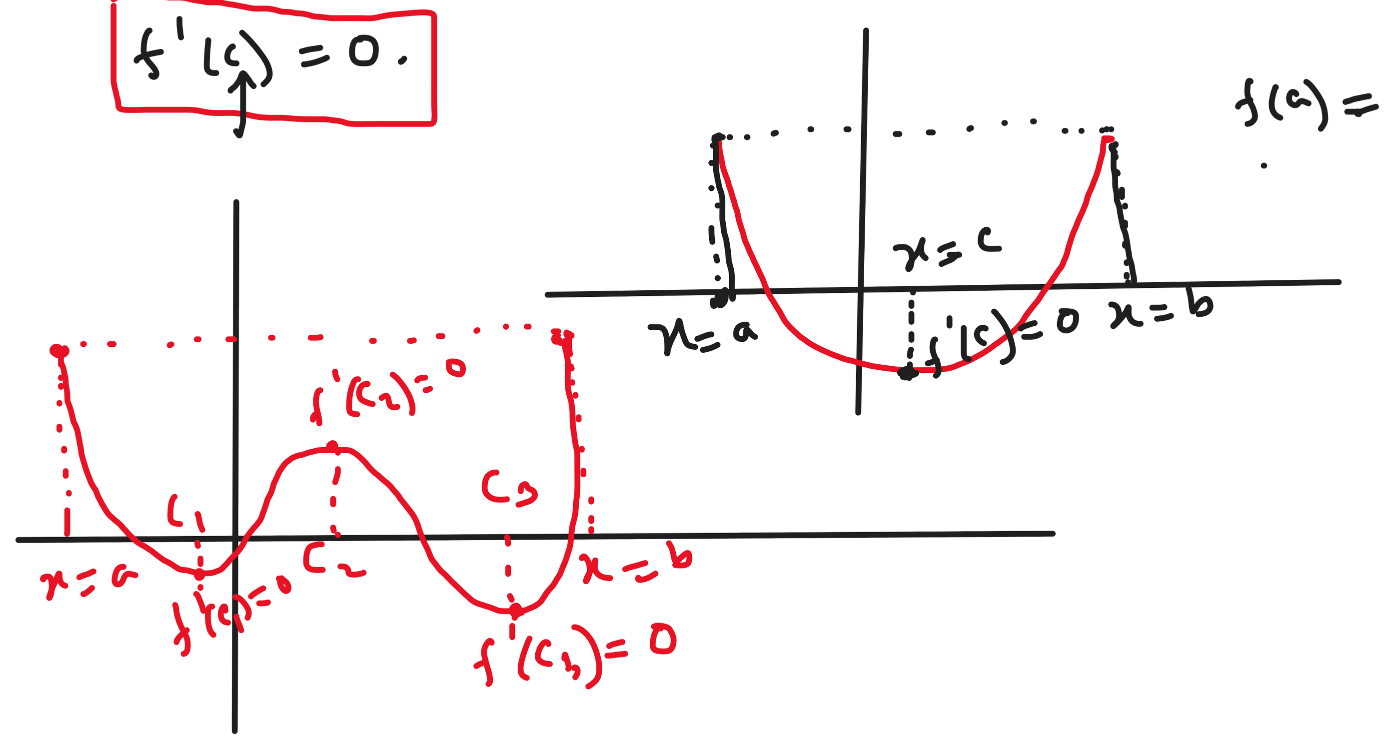
# Rolle's theorem

**Statement:** Let $f$ be differentiable on $(a,b)$ and continuous on $[a,b]$. If $f(a) = f(b)$, then there is at least one number $c$ in $(a,b)$ such that $\boxed{f'(c) = 0.}$




**Ex:** Verify the hypothesis and conclusions of Rolle's theorem for $\boxed{f(x) = \sin x}$ on $(0, 2\pi)$.

**Sol^n:** Since $f(x) = \sin x$ is continuous and differentiable everywhere, it is differentiable on $(0, 2\pi)$ and continuous on $[0, 2\pi]$.

Here, $f(0) = \sin 0 = 0$ and $f(2\pi) = \sin 2\pi = 0$

$\therefore f(0) = f(2\pi)$

Thus, by the Rolle's theorem, we can say there is at least one number $c \in (0, 2\pi)$ such that $\boxed{f'(c) = 0}$.

$$\boxed{\begin{aligned} &\cos x = 0 \\ \Rightarrow\ &x = \frac{(2n+1)\pi}{2} \\ &n \in \mathbb{Z} \end{aligned}}$$

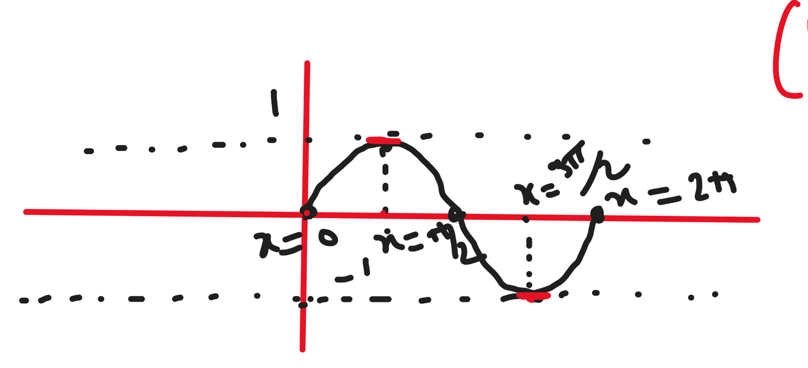
$f'(x) = \cos x$ $\qquad \Rightarrow \cos c = 0$

$f'(c) = \cos c$ $\qquad \Rightarrow c = \frac{\pi}{2}, \frac{3\pi}{2}$

$\therefore c = \frac{\pi}{2}, \frac{3\pi}{2} \in (0, 2\pi)$

(Verified)

$f(x) = \sin x$
$(0, 2\pi)$

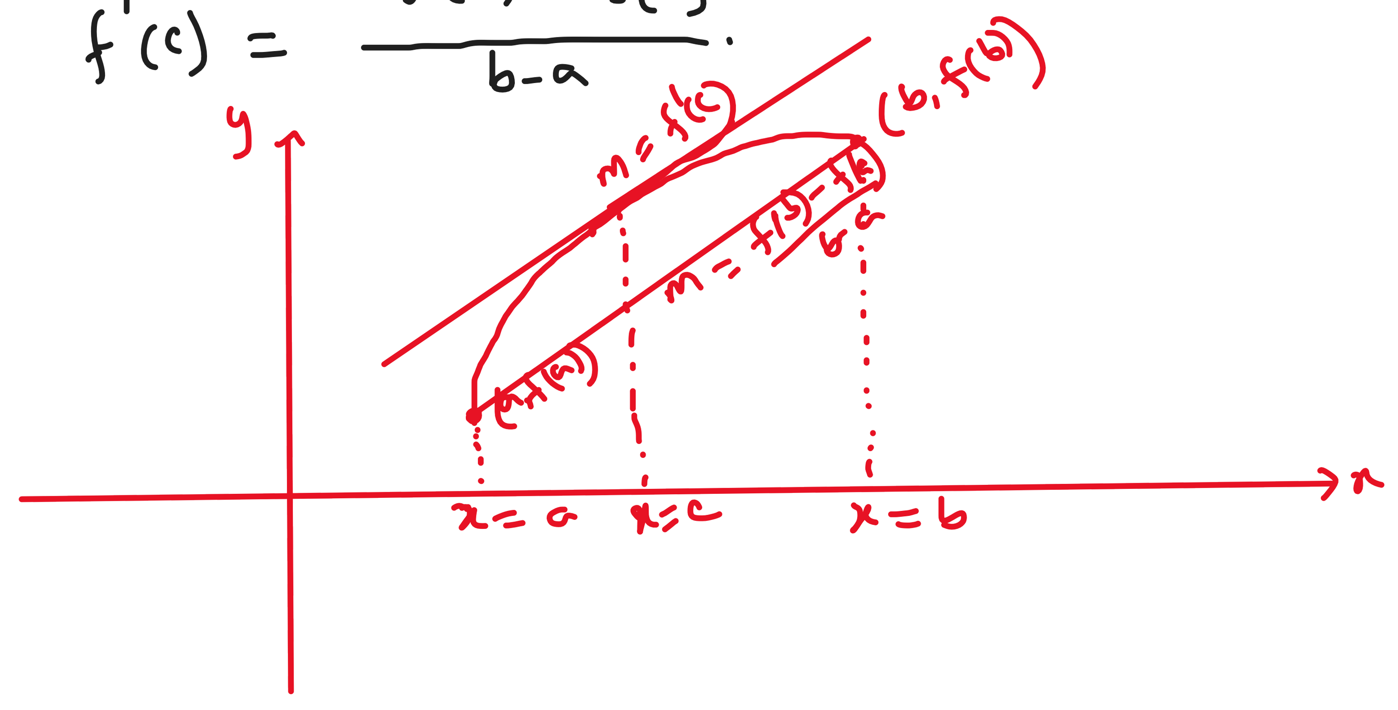



$f(x) = \underline{g(x)}\ \underline{h(x)}$ , $f(x) = g(x) - h(x)$

# Mean Value Theorem

**Statement:** Let $f$ be differentiable on $(a,b)$ and continuous on $[a,b]$. Then, there is at least one number $c$ in $(a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$




**Ex:** Verify the Mean Value theorem for $f(x) = e^{-x} \sin x$ on $(0, \pi)$.

**Sol^n:** Since $f(x) = e^{-x} \sin x$ is continuous on $[0, \pi]$ and differentiable on $(0, \pi)$. So, by Mean Value theorem, there is at least one number $c$ in $(0, \pi)$ such that,

$$f'(c) = \frac{f(\pi) - f(0)}{\pi - 0} \quad \text{——— (*)}$$

$f(x) = e^{-x} \sin x$

$f'(x) = -e^{-x} \sin x + e^{-x} \cos x$

$\therefore f'(c) = e^{-c}(-\sin c + \cos c)$

$f(0) = e^{-0} \sin 0 = 0$

$f(\pi) = e^{-\pi} \sin \pi = 0$

From (*),

$e^{-c}(-\sin c + \cos c) = \frac{0 - 0}{\pi - 0} = 0$

$\Rightarrow e^{-c} \neq 0, \quad -\sin c + \cos c = 0$

$\Rightarrow \cos c = \sin c$

$\Rightarrow 1 = \frac{\sin c}{\cos c} = \tan c$

$\Rightarrow \tan c = 1$

$\Rightarrow c = \pi/4 \in (0, \pi)$

(Verified)